

A dimension-independent commutator bound via trace-mass splitting

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Abstract

We prove a dimension-independent operator-norm bound for a single commutator representing a complex trace-zero matrix. The argument splits according to the trace norm of the rotated Hermitian parts. In the high-mass case, a numerical-range argument produces an invertible off-diagonal compression of proportional rank. A matrix Riccati equation solves an identity-corner core, and finitely many similarities absorb the remaining diagonal blocks. In the low-mass case, spectral rearrangement, simultaneous hollowisation, and the Marcus–Spielman–Srivastava vector theorem give an equal-rank, trace-preserving paving with compression norm of order R^{-1} . A finite-block Sylvester construction then closes a strong induction. All commutator factors act on the original space. We give explicit, very large constants and a complete Lean formalization of the theorem in the Euclidean operator norm.

Formal verification. The companion Lean project proves Theorem 1.1, including the dimension-independent quantifiers and the Euclidean operator norm. The final kernel dependency audit returns only Lean’s three standard logical axioms; see Appendix A. Published inputs are identified at their points of use. This anonymous manuscript has not been submitted to arXiv.

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1 Introduction and main theorem

For $A \in M_n(\mathbb{C})$, write $\|A\|$ for its operator norm on Euclidean space and $[B, C] = BC - CB$. The quantitative commutator problem asks how economically one can represent a trace-zero matrix by a single commutator. Johnson, Ozawa, and Schechtman proved that, for each $\varepsilon > 0$, a bound $K_\varepsilon n^\varepsilon \|A\|$ is possible; their first factor can be chosen normal [3]. Ravichandran and Srivastava obtained the bound $300 \exp(9\sqrt{\log n}) \|A\|$ [6, Corollary 2]. The theorem considered here allows both factors to be arbitrary complex matrices.

Theorem 1.1 (Main theorem). *There is an absolute finite constant $K > 0$ such that, for every integer $n \geq 1$ and every $A \in M_n(\mathbb{C})$ with $\text{tr } A = 0$, there exist $B, C \in M_n(\mathbb{C})$ satisfying*

$$A = [B, C], \quad \|B\| \|C\| \leq K \|A\|. \quad (1.1)$$

The same K works for every n .

The construction has two branches. Set

$$s = 8192, \quad R = s^2 = 2^{26}, \quad b = 8sR = 2^{42}, \quad t_0 = 2/R. \quad (1.2)$$

A matrix with large trace norm in every rotated Hermitian direction is handled directly, with a constant $K_H(t_0)$. Otherwise it admits R trace-zero compressions of norm at most $319/R$ times its own norm. Reassembly costs a factor $8s$ on the largest child budget, and

$$8s \frac{319}{R} = \frac{2552}{8192} < \frac{1}{2}.$$

This strict inequality is the quantitative reason the induction closes.

1.1 Conventions and elementary norm estimates

All spaces are finite-dimensional complex Hilbert spaces. A compression to a subspace V means $P_V A|_V$; no invariance of V is implicit. Rectangular blocks carry their operator norms. The trace is unnormalized, and $|H| = (H^2)^{1/2}$ for Hermitian H . We use

$$\text{Re } A = (A + A^*)/2, \quad \text{Im } A = (A - A^*)/(2i), \quad \chi(S) = \|S\| \|S^{-1}\|.$$

Unitary conjugation preserves the norm and trace. A similarity multiplies a commutator product budget by at most $\chi(S)^2$.

We shall use a basic block estimate repeatedly. If an operator $T = (T_{ij})$ has zero diagonal in an orthogonal decomposition with k blocks and $\|T_{ij}\| \leq c$ for $i \neq j$, then

$$\|T\| \leq c(k-1). \quad (1.3)$$

Indeed, for $x = (x_j)$, the vector of component norms of Tx is bounded coordinatewise by $c(\mathbf{1}\mathbf{1}^\top - I)(\|x_j\|)_j$. The scalar matrix has Euclidean operator norm $k-1$ when $k \geq 2$; the case $k=1$ is immediate. Also, the norm of a block diagonal operator is the largest block norm, and the norm of a general block matrix is at most the sum of its block norms.

1.2 External inputs

Besides finite-dimensional spectral theory, the Toeplitz–Hausdorff theorem, compact convexity, and the Banach fixed-point theorem, the proof uses three specific published inputs: the two-dimensional Steinitz rearrangement theorem, the simultaneous hollowisation theorem of Damm and Faßbender, and the independent-vector theorem of Marcus, Spielman, and Srivastava. Their exact forms are stated in Section 6. The two-commutator decomposition in the next section is elementary and allows its first factors to depend on the input matrix.

2 An elementary two-commutator decomposition

Lemma 2.1. *Let $m \geq 1$. If $G \in M_m(\mathbb{C})$ is Hermitian and $\text{tr } G = 0$, there are a unitary u and a matrix V such that $G = [u, V]$ and $\|V\| \leq \|G\|$.*

Proof. For $m = 1$, take $u = I$ and $V = 0$. Otherwise diagonalize G unitarily and denote its real eigenvalues by $\lambda_1, \dots, \lambda_m$. They sum to zero and lie in $[-a, a]$, where $a = \|G\|$. They can be ordered so that every partial sum $s_j = \sum_{i=1}^j \lambda_i$ belongs to $[-a, a]$. To see this, if the current sum is positive, choose a remaining negative term; such a term exists because the remaining terms have negative sum. If the current sum is negative, choose a positive term. At zero choose any remaining term. In each case the new sum stays in $[-a, a]$. This constructs the required ordering, including zero terms.

In this ordered eigenbasis, let

$$u = \sum_{j=1}^{m-1} E_{j,j+1} + E_{m,1}, \quad V = \sum_{j=1}^{m-1} s_j E_{j+1,j}.$$

The first matrix is unitary, $\|V\| = \max_{j < m} |s_j| \leq a$, and multiplication gives

$$uV = \text{diag}(s_1, \dots, s_{m-1}, 0), \quad Vu = \text{diag}(0, s_1, \dots, s_{m-1}).$$

Their difference is $\text{diag}(\lambda_1, \dots, \lambda_m)$, since $s_m = 0$. Conjugate both factors back to the original basis. $\square$

Corollary 2.2. *Let $m \geq 1$. Every trace-zero $H \in M_m(\mathbb{C})$ has a representation*

$$H = [u, V] + [k, T], \quad u, k \text{ unitary}, \quad \max(\|V\|, \|T\|) \leq \|H\|. \quad (2.1)$$

Proof. Write $H = H_1 + iH_2$ with H_1, H_2 Hermitian, trace zero, and of norm at most $\|H\|$. Apply Lemma 2.1 to obtain $H_1 = [u, V]$ and $H_2 = [k, W]$, and set $T = iW$. $\square$

Remark 2.3. The quantifiers in (2.1) are $\forall H \exists u, k, V, T$. We do not assert that one pair u, k works uniformly for all H in a fixed dimension, or that the factors vary continuously with H . Neither property is needed below: the four factors are chosen once before the Riccati equation is solved.

3 Solving an identity-corner core

For $a \geq 1$, define the following scalar quantities:

$$\begin{aligned} p &= 2a, & d_0 &= 5a, & q &= 5a + 1, & d_1 &= 14a + 1, \\ L_0 &= q^2 + (2a + 1)q + d_1, & L_1 &= 2q + 2a + 1, & \lambda &= 2 \max(L_0, L_1), \\ P(a) &= (1 + q)^4(\lambda + 3)((\lambda + 5)p + 2). \end{aligned} \quad (3.1)$$

All of these are finite and nondecreasing in a .

Proposition 3.1. *Let $m \geq 1$, $a \geq 1$, and*

$$M = \begin{pmatrix} A & I_m \\ F & D \end{pmatrix}, \quad \operatorname{tr} M = 0, \quad \|M\| \leq a.$$

Then M is a single commutator in $M_{2m}(\mathbb{C})$ with factor norm product at most $P(a)$.

Proof. Since $\operatorname{tr}(A + D) = 0$, Corollary 2.2 supplies fixed factors such that

$$A + D = [u, V] + [k, T], \quad \|u\| = \|k\| = 1, \quad \|V\|, \|T\| \leq p.$$

Set

$$U = \lambda I + u, \quad D_0 = [u, V] - A, \quad D_1 = [k, V] + D_0 u - k + [u, Tk] - F. \quad (3.2)$$

The block compression bounds imply $\|A\|, \|D\|, \|F\| \leq a$, whence

$$\|D_0\| \leq 2p + a = d_0, \quad \|D_1\| \leq 2p + d_0 + 1 + 2p + a = d_1.$$

On the closed ball $\mathcal{B} = \{Q : \|Q - D_0\| \leq 1\}$ consider

$$\Psi(Q) = D_0 + \lambda^{-1}(Q^2 - DQ + Q(A - u) + D_1). \quad (3.3)$$

Every $Q \in \mathcal{B}$ has norm at most q , so

$$\|\Psi(Q) - D_0\| \leq L_0/\lambda \leq \frac{1}{2}.$$

For $Q, Q' \in \mathcal{B}$, the noncommutative identity $Q^2 - Q'^2 = (Q - Q')Q + Q'(Q - Q')$ gives

$$\|\Psi(Q) - \Psi(Q')\| \leq \frac{2q + 2a + 1}{\lambda} \|Q - Q'\| \leq \frac{1}{2} \|Q - Q'\|.$$

The ball is complete. The Banach fixed-point theorem gives an exact fixed point $Q \in \mathcal{B}$.

Define

$$B_c = \begin{pmatrix} U & I \\ k & 0 \end{pmatrix}, \quad C_c = \begin{pmatrix} V & T \\ A + Q - [u, V] + Tk & V + I - UT \end{pmatrix}, \quad S = \begin{pmatrix} I & 0 \\ Q & I \end{pmatrix}. \quad (3.4)$$

To verify the order of all factors, put

$$F_0 = [k, V] + D_0 U - k + [u, Tk] = \lambda D_0 + D_1 + F.$$

Direct multiplication, using $A + D = [u, V] + [k, T]$, yields

$$\begin{aligned} [B_c, C_c] &= \begin{pmatrix} A + Q & I \\ F_0 - QU & D - Q \end{pmatrix}, \\ S^{-1}MS &= \begin{pmatrix} A + Q & I \\ F + DQ - QA - Q^2 & D - Q \end{pmatrix}. \end{aligned} \quad (3.5)$$

The bottom-left blocks agree precisely when

$$Q^2 - DQ + Q(A - U) + F_0 - F = 0,$$

which is equivalent to (3.3). Consequently

$$M = [SB_c S^{-1}, SC_c S^{-1}]. \quad (3.6)$$

It remains to bound the factors. The block norm sum gives $\|B_c\| \leq \lambda + 3$. The bottom-left block of C_c equals $Q - D_0 + Tk$, of norm at most $1 + p$, and the bottom-right block has norm at most $1 + (\lambda + 2)p$. Thus

$$\|C_c\| \leq (\lambda + 5)p + 2, \quad \|S\|, \|S^{-1}\| \leq 1 + q.$$

Equation (3.6) now gives the claimed $P(a)$. $\square$

4 Absorbing a bounded number of remaining blocks

Lemma 4.1 (Sylvester equation with separated centers). *Let U and V act on possibly different Hilbert spaces, and let $z, w \in \mathbb{C}$. If $\eta = \|U\| + \|V\| < |z - w|$, then, for every rectangular Y , the equation*

$$(z - w)X + UX - XV = Y$$

has a solution with $\|X\| \leq \|Y\| / (|z - w| - \eta)$.

Proof. On the space of rectangular operators with its operator norm, the map $\mathcal{E}(X) = UX - XV$ has norm at most η . The inverse of $(z - w)I + \mathcal{E}$ is the norm-convergent Neumann series

$$\frac{1}{z - w} \sum_{j=0}^{\infty} \left(-\frac{\mathcal{E}}{z - w} \right)^j,$$

whose norm is at most $(|z - w| - \eta)^{-1}$. □

Proposition 4.2. *Let a trace-zero matrix A have an orthogonal block decomposition of sizes $r, r, s_3, \dots, s_b$, where $b \geq 2$, $r \geq 1$, and $1 \leq s_j \leq r$. Suppose $A_{12} = I_r$ and $\|A\| \leq a$, with $a \geq 1$. Put*

$$\beta(x) = x(x + 2)^2, \quad f(x) = \beta(x)(1 + \beta(x))^2, \quad \hat{a} = f^{\circ(b-2)}(a), \quad k = b - 1. \quad (4.1)$$

The function f is nondecreasing and satisfies $f(x) \geq x$ for $x \geq 1$. Then A has a same-dimensional single-commutator representation of cost at most

$$(\hat{a}/a)^2 k (4P(\hat{a}) + 2k\hat{a}). \quad (4.2)$$

Proof. Write $E_{ij}(X)$ for the operator supported in block (i, j) . Fix $j \geq 3$ and an isometric embedding $J : \mathbb{C}^{s_j} \rightarrow \mathbb{C}^r$, so that $J^*J = I$. First use

$$X = J - A_{1j}, \quad S_1 = I + E_{2j}(X), \quad A' = S_1^{-1}AS_1.$$

Since $E_{2j}(X)^2 = 0$, the inverse of S_1 is $I - E_{2j}(X)$. The first block row is unchanged by the left multiplication, and therefore

$$A'_{1j} = A_{1j} + A_{12}X = J, \quad A'_{12} = I.$$

Only diagonal blocks 2 and j can change. Indeed, in the expansion $S_1^{-1}AS_1 = A + AE_{2j}(X) - E_{2j}(X)A - E_{2j}(X)AE_{2j}(X)$, the last term is supported in block $(2, j)$.

Next set

$$Y = A'_{jj}J^*, \quad S_2 = I + E_{j1}(Y), \quad A'' = S_2^{-1}A'S_2.$$

Right multiplication changes only the first block column, while left multiplication subtracts Y times the first block row from row j . Hence

$$A''_{jj} = A'_{jj} - YA'_{1j} = 0, \quad A''_{12} = I. \quad (4.3)$$

Only diagonal blocks 1 and j can change in this step. Previously eliminated outside diagonal blocks remain zero. Applying these two shears for every $j = 3, \dots, b$ eliminates all outside diagonal blocks while preserving the identity corner.

For the quantitative bounds, suppose the current norm is at most $x \geq 1$. Then $\|X\| \leq x + 1$ and

$$\chi(S_1) \leq (x + 2)^2, \quad \|A'\| \leq \beta(x).$$

Since $\|Y\| \leq \beta(x)$,

$$\chi(S_2) \leq (1 + \beta(x))^2, \quad \|A''\| \leq f(x).$$

The product S of all these shears consequently satisfies

$$\hat{A} = S^{-1}AS, \quad \|\hat{A}\| \leq \hat{a}, \quad \chi(S) \leq \prod_{j=0}^{b-3} \frac{f^{\circ(j+1)}(a)}{f^{\circ j}(a)} = \hat{a}/a. \quad (4.4)$$

Empty products have value one when $b = 2$.

Merge the first two blocks into one core M . Since every other diagonal block is zero and similarity preserves trace, $\text{tr } M = 0$. Proposition 3.1 gives $M = [B_0, C_0]$ with product at most $P(\hat{a})$. Its identity corner ensures $M \neq 0$. Rescale the two factors inversely to arrange

$$\|B_0\| = 1/4, \quad \|C_0\| \leq 4P(\hat{a}).$$

There are now k blocks. Let

$$\hat{B} = \text{diag}(B_0, I, 2I, \dots, (k-1)I), \quad \hat{C}_{11} = C_0, \quad \hat{C}_{jj} = 0 \ (j > 1).$$

For distinct blocks solve $\hat{B}_{ii}\hat{C}_{ij} - \hat{C}_{ij}\hat{B}_{jj} = \hat{A}_{ij}$ using Lemma 4.1. The centers are distinct integers and the only non-scalar perturbation has norm $1/4$. Thus $\|\hat{C}_{ij}\| \leq 2\hat{a}$ and

$$\|\hat{B}\| \leq k, \quad \|\hat{C}\| \leq 4P(\hat{a}) + 2(k-1)\hat{a}$$

by (1.3). These choices give $[\hat{B}, \hat{C}] = \hat{A}$. Conjugating back and applying (4.4) proves (4.2). $\square$

5 The high trace-mass branch

For $0 < t \leq 1$, say that $A \in M_n(\mathbb{C})$ satisfies $\text{HM}(t)$ if

$$\text{tr } |\text{Re}(e^{i\theta}A)| \geq tn \|A\| \quad \text{for every } \theta \in \mathbb{R}. \quad (5.1)$$

Lemma 5.1 (A disk in compressed numerical ranges). *Suppose $\text{tr } A = 0$, $\|A\| = 1$, and A satisfies $\text{HM}(t)$. If W has codimension $d \leq tn/4$, then the numerical range of $T = P_W A|_W$ contains the closed disk of radius $\rho = t/4$ centered at zero.*

Proof. For every θ , the Hermitian matrix $H_\theta = \text{Re}(e^{i\theta}A)$ has norm at most one and trace zero. Its positive eigenvalues therefore have sum at least $tn/2$. Write the eigenvalues in decreasing order. Since $d < tn/2$, at least $d+1$ eigenvalues are positive; moreover

$$tn/2 \leq d + (n-d)\lambda_{d+1}(H_\theta).$$

It follows that

$$\lambda_{d+1}(H_\theta) \geq \frac{tn/2 - d}{n - d} \geq t/4.$$

The min-max principle yields $\lambda_{\max}(P_W H_\theta|_W) \geq t/4$. The numerical range is compact and convex by the Toeplitz-Hausdorff theorem. Its support in each direction is this largest eigenvalue, so it contains the stated disk. Equivalently, use the supporting-half-plane description of the ordinary numerical range; see [4, Section 2]. $\square$

Lemma 5.2 (A neutral vector with a large image). *If the numerical range of T contains both ρ and $-\rho$, where $\rho > 0$, then there is a unit vector v such that*

$$v^*Tv = 0, \quad \|Tv\| \geq \rho. \quad (5.2)$$

Proof. Choose unit vectors x_+, x_- with expectations $\rho, -\rho$. The density matrix $D_* = (x_+x_+^* + x_-x_-^*)/2$ has trace one, satisfies $\text{tr}(TD_*) = 0$, and obeys $\text{tr}(T^*TD_*) \geq \rho^2$ by Cauchy–Schwarz.

On the compact convex set

$$\mathcal{D} = \{D \geq 0 : \text{tr } D = 1, \text{tr}((\text{Re } T)D) = 0, \text{tr}((\text{Im } T)D) = 0\},$$

maximize the real linear functional $\text{tr}(T^*TD)$. The set of maximizers is a nonempty compact face, so it has an extreme point that is also extreme in $\mathcal{D}$. Every extreme point of $\mathcal{D}$ has rank one. In fact, if an extreme D had rank $s \geq 2$, the Hermitian operators on its support would form a real vector space of dimension $s^2 \geq 4$. The three displayed real linear constraints admit a nonzero supported Hermitian perturbation E in their common kernel. Because D is strictly positive on its support, both $D + \eta E$ and $D - \eta E$ belong to $\mathcal{D}$ for sufficiently small $\eta > 0$, a contradiction. The maximizing extreme point is thus vv^* , and its objective value is at least that of D_* . $\square$

Proposition 5.3. *Every trace-zero matrix satisfying $\text{HM}(t)$ has a same-dimensional single commutator representation with factor product at most $K_H(t) \|A\|$, where*

$$\begin{aligned} L &= \lceil 16/t \rceil, & a_* &= f^{\circ L}(4/t), & k_* &= L + 1, \\ K_H(t) &= a_*^2 k_* (4P(a_*) + 2k_* a_*) < \infty. \end{aligned} \quad (5.3)$$

Proof. The zero matrix is immediate. Normalize $\|A\| = 1$, and set $\rho = t/4$ and $r = \lceil tn/16 \rceil$. We construct orthonormal $v_1, \dots, v_r$ such that all $v_i^* Av_j$ vanish and the vectors Av_i are pairwise orthogonal with norms at least ρ .

After selecting $v_1, \dots, v_j$, work in

$$W_j = \text{span}\{v_i, Av_i, A^*v_i, A^*Av_i : 1 \leq i \leq j\}^\perp.$$

For $0 \leq j \leq r - 1$ we have $\text{codim } W_j \leq 4j < tn/4$. Lemmas 5.1 and 5.2, applied to $P_{W_j}A|_{W_j}$, give a unit $v_{j+1} \in W_j$ with

$$v_{j+1}^* Av_{j+1} = 0, \quad \|Av_{j+1}\| \geq \|P_{W_j}Av_{j+1}\| \geq \rho.$$

Orthogonality to v_i preserves orthonormality; orthogonality to Av_i and A^*v_i makes both mixed matrix coefficients vanish; orthogonality to A^*Av_i makes Av_{j+1} perpendicular to Av_i . This proves the induction.

Let $V = \text{span}\{v_i\}$ and $W = A(V)$. The construction implies $V \perp W$ and $\dim V = \dim W = r$; in particular $2r \leq n$. In the ordered bases $w_i = Av_i / \|Av_i\|$ of W , followed by v_i of V , the $(1, 2)$ block of A is

$$D = \text{diag}(\|Av_1\|, \dots, \|Av_r\|), \quad \rho I \leq D \leq I.$$

The similarity $S_0 = \text{diag}(I_r, D^{-1}, I_{\text{rest}})$ has condition number at most $4/t$. In $S_0^{-1}AS_0$, the $(1, 2)$ block equals I_r and the norm is at most $a_0 = 4/t$.

Partition the remaining space into blocks of positive size at most r . Since $r \geq tn/16$, the number of outside blocks is at most $L = \lceil 16/t \rceil$. Apply Proposition 4.2. The iterated norm bound is at most a_* , and after including S_0 the total similarity condition number is at most $(4/t)(a_*/a_0) = a_*$. There are at most k_* blocks after merging the core. Its proof gives precisely the bound (5.3). Finally undo the scalar normalization. $\square$

6 The low trace-mass branch

The key point of this branch is a complete paving whose individual traces are exactly zero and whose ranks are equal.

6.1 Rearrangement, hollowisation, and vector selection

We use the following form of the Steinitz theorem: if $u_1, \dots, u_N \in \mathbb{R}^2$ sum to zero and $\|u_i\|_\infty \leq 1$, they can be reordered so that every partial sum has ℓ_∞ norm at most 2. This is the $d = 2$ instance of [1, Theorem 1.2].

Damm and Faßbender prove that three trace-zero Hermitian matrices can be simultaneously conjugated by a unitary so that the first two have zero diagonal and the third has at most two nonzero diagonal entries [2, Proposition 13(b)]. The last two entries need not vanish.

Lemma 6.1. *Let H, G be Hermitian operators on $\mathbb{C}^R$, with $R \geq 2$, and let $E \geq 0$. There is an orthonormal basis $v_1, \dots, v_R$ such that*

$$v_a^* H v_a = \frac{\text{tr } H}{R}, \quad v_a^* G v_a = \frac{\text{tr } G}{R}, \quad v_a^* E v_a \leq \frac{2 \text{tr } E}{R} \quad (1 \leq a \leq R). \quad (6.1)$$

Proof. Subtract the scalar mean from each of the three matrices and apply the cited simultaneous hollowisation theorem. In the resulting basis the first two means are attained at every vector. All but possibly two of the diagonal entries of E equal $\mu = \text{tr } E/R$. The remaining two entries have sum 2μ and are nonnegative; each is therefore at most 2μ . This argument also covers $R = 2$. $\square$

We also use the following theorem of Marcus, Spielman, and Srivastava [5, Theorem 1.4]. If independent, finitely supported random vectors x_i in a complex Hilbert space satisfy

$$\sum_i \mathbb{E} x_i x_i^* = I, \quad \mathbb{E} \|x_i\|^2 \leq \varepsilon, \quad (6.2)$$

where $\varepsilon > 0$, some outcome satisfies

$$\left\| \sum_i x_i x_i^* \right\| \leq (1 + \sqrt{\varepsilon})^2. \quad (6.3)$$

The same conclusion holds if the covariance sum is merely at most I . Indeed, spectrally decompose its positive semidefinite deficit and split each eigenvalue into finitely many pieces at most ε . Add the corresponding deterministic vectors, apply the theorem, and then discard their positive semidefinite contributions. No upper restriction on ε is required.

6.2 Paving with exact quotas in every group

Lemma 6.2. *Let $D \geq 0$ and let h be a nonnegative integer. Suppose vectors are indexed by finitely many groups of size $R = 2^h$ and satisfy*

$$\sum_i f_i f_i^* \leq I, \quad \|f_i\|^2 \leq D/R.$$

Their indices can be partitioned into R transversals, each containing exactly one index from each group, such that every transversal J satisfies

$$\left\| \sum_{i \in J} f_i f_i^* \right\| \leq \frac{C_D}{R}, \quad C_D = [1 + 2(\sqrt{2} + 1)\sqrt{D}]^2. \quad (6.4)$$

Proof. If $D = 0$ all the vectors vanish, so any transversal partition works. Assume $D > 0$ and write $\delta = D/R$. Consider a node containing equally many indices from each group, an even number per group, whose frame operator is at most LI . Pair the indices within each group. For a pair of vectors a, b , choose independently and uniformly one of the four vectors

$$\sqrt{2}(a \oplus b), \quad \sqrt{2}(a \oplus (-b)), \quad \sqrt{2}(b \oplus a), \quad \sqrt{2}(b \oplus (-a)). \quad (6.5)$$

For the random vector w so obtained, the off-diagonal expected covariance blocks cancel. On summing over the pairs,

$$\sum \mathbb{E} w w^* = \text{diag} \left(\sum_i f_i f_i^*, \sum_i f_i f_i^* \right) \leq LI, \quad \mathbb{E} \|w\|^2 \leq 4\delta.$$

When $L > 0$, apply (6.3) to $w/\sqrt{L}$ with $\varepsilon = 4\delta/L$. Some outcome has

$$\left\| \sum w w^* \right\| \leq (\sqrt{L} + 2\sqrt{\delta})^2.$$

Its two diagonal blocks equal twice the frame operators of two complementary halves, each containing one member of each pair. Consequently both children have frame operator bounded by

$$L_{\text{next}} I, \quad L_{\text{next}} = \frac{1}{2}(\sqrt{L} + 2\sqrt{\delta})^2. \quad (6.6)$$

If $L = 0$, the vectors at the node are zero and can be divided arbitrarily.

Repeat this operation at every node for h levels, starting with $L_0 = 1$. At depth j every node contains $R/2^j$ indices from each original group. Thus the leaves are disjoint transversals covering all indices. Taking square roots in (6.6) and summing the geometric progression gives

$$\sqrt{L_h} \leq 2^{-h/2} + \sqrt{2\delta} \sum_{j=0}^{h-1} 2^{-j/2} = R^{-1/2} + 2(\sqrt{2} + 1)\sqrt{D/R}(1 - R^{-1/2}),$$

which implies (6.4). The construction partitions indices, so repeated or zero vector values cause no ambiguity. $\square$

6.3 Trace-preserving small compressions

Proposition 6.3. *Let $h \geq 1$ and $k \geq 1$ be integers, set $R = 2^h$ and $n = kR$, and suppose $\text{tr } A = 0$ and, for some $\theta \in \mathbb{R}$,*

$$\text{tr} | \text{Re}(e^{i\theta} A) | \leq 2k \|A\|. \quad (6.7)$$

There are R mutually orthogonal rank- k projections P_ℓ summing to I such that

$$\text{tr}(P_\ell A P_\ell) = 0, \quad \|P_\ell A P_\ell\| \leq \frac{319}{R} \|A\|. \quad (6.8)$$

Proof. The case $A = 0$ is immediate. Normalize and rotate so that $\|A\| = 1$ and $A = H + iG$, where H, G are trace-zero Hermitian matrices and

$$E = |H| \leq I, \quad \text{tr } E \leq 2k.$$

Choose an orthonormal eigenbasis e_i of G , with eigenvalues λ_i . Set

$$d_i = e_i^* E e_i, \quad \mu = \frac{\text{tr } E}{kR}.$$

Since $|\lambda_i| \leq 1$, $0 \leq d_i, \mu \leq 1$, the real vectors $(\lambda_i, d_i - \mu)$ sum to zero and have ℓ_∞ norm at most one. Apply Steinitz rearrangement and divide the reordered list into k consecutive groups of size R . Each group sum is a difference of two partial sums. The corresponding orthogonal spectral subspaces V_j therefore satisfy

$$|\operatorname{tr} G_j| \leq 4, \quad \operatorname{tr} E_j \leq R\mu + 4 \leq 6, \quad (6.9)$$

where subscripts denote compression to V_j . The spaces V_j reduce G .

Apply Lemma 6.1 to H_j, G_j, E_j in each V_j . This gives an orthonormal basis $v_{j,a}$ with

$$v_{j,a}^* H v_{j,a} = \operatorname{tr} H_j / R, \quad v_{j,a}^* G v_{j,a} = \operatorname{tr} G_j / R, \quad v_{j,a}^* E v_{j,a} \leq 12/R. \quad (6.10)$$

For $f_{j,a} = E^{1/2} v_{j,a}$ we have

$$\sum_{j,a} f_{j,a} f_{j,a}^* = E \leq I, \quad \|f_{j,a}\|^2 \leq 12/R.$$

Lemma 6.2 with $D = 12$ gives R transversals. Let P_ℓ project onto the span of the original basis vectors in transversal ℓ . These projections are orthogonal, sum to I , and have rank k .

Equation (6.10) implies that the trace of each compression of A is

$$\frac{1}{R} \sum_j (\operatorname{tr} H_j + i \operatorname{tr} G_j) = \frac{\operatorname{tr} A}{R} = 0.$$

Since distinct V_j reduce G , the compression $P_\ell G P_\ell$ is diagonal in its selected basis, and (6.9) gives $\|P_\ell G P_\ell\| \leq 4/R$. Furthermore

$$\|P_\ell E P_\ell\| = \|E^{1/2} P_\ell E^{1/2}\| = \left\| \sum_{(j,a) \in \ell} f_{j,a} f_{j,a}^* \right\| \leq C_{12}/R.$$

The first equality follows from $\|X^* X\| = \|X X^*\|$ for $X = E^{1/2} P_\ell$. Since $-E \leq H \leq E$, order and the Hermitian norm formula imply $\|P_\ell H P_\ell\| \leq C_{12}/R$. Thus

$$\|P_\ell A P_\ell\| \leq (C_{12} + 4)/R < 319/R.$$

For completeness, $\sqrt{2} < 1.415$ and $\sqrt{12} < 3.465$ give $1 + 2(\sqrt{2} + 1)\sqrt{12} < 17.736$, whose square is less than 315. Undo the rotation and normalization to obtain (6.8). $\square$

7 Finite-block assembly and the global induction

Lemma 7.1. *With s, R, b as in (1.2), suppose A is decomposed into $1 \leq r \leq 2R - 1$ nonempty orthogonal blocks and every diagonal block has a representation $A_{ii} = [U_i, V_i]$ of cost $p_i = \|U_i\| \|V_i\|$. Then $A = [B, C]$ in its original dimension, with*

$$\|B\| \|C\| \leq 8s \max_i p_i + b \|A\|. \quad (7.1)$$

Proof. For a nonzero diagonal block, inversely rescale its two factors to make $\|U_i\| = 1/4$ and $\|V_i\| = 4p_i$. For a zero block use two zero factors; this only decreases its cost. Choose distinct centers z_i from the square grid

$$\{x + iy : x, y \in \{-s + \frac{1}{2}, -s + \frac{3}{2}, \dots, s - \frac{1}{2}\}\}.$$

There are $4s^2 = 4R$ centers, with pairwise distance at least one. Define

$$B = \text{diag}(z_i I + U_i), \quad C_{ii} = V_i.$$

Since $|z_i| \leq \sqrt{2}(s - 1/2)$, we have $\|B\| < 2s$. For each $i \neq j$ solve

$$(z_i - z_j)C_{ij} + U_i C_{ij} - C_{ij} U_j = A_{ij}.$$

Lemma 4.1 applies with perturbation norm at most $1/2$, giving $\|C_{ij}\| \leq 2\|A_{ij}\| \leq 2\|A\|$. The block equations give $[B, C] = A$, and (1.3) implies

$$\|C\| \leq 4 \max_i p_i + 2(r-1)\|A\|.$$

Multiply by $2s$ and use $r-1 \leq 2R$ to obtain (7.1). No normality assumption on any U_i is used. $\square$

Proof of Theorem 1.1. Let $K_H = K_H(t_0)$ be defined by (5.3), and set

$$K = \max\{8sK_H + b, 2b\} = \max\{65536K_H + 2^{42}, 2^{43}\}. \quad (7.2)$$

This is a finite absolute constant. We argue by strong induction on n . The zero matrix has the required representation, which also covers dimension one. For a nonzero A , put $a = \|A\| > 0$.

First choose an orthonormal basis in which A has zero diagonal. Here is a short justification. The average of the diagonal expectations in any orthonormal basis is zero. By convexity of the numerical range, zero is a vector expectation. Choose a unit vector with this expectation; the compression to its orthogonal complement still has trace zero. Repeat in the smaller dimension. This also follows from [2, Corollary 14].

If $n < R$, use the n zero singleton diagonal blocks in Lemma 7.1; the cost is at most $ba \leq Ka$. If $n \geq R$, write

$$n = kR + q, \quad k \geq 1, \quad 0 \leq q < R.$$

Retain $N = kR$ vectors of the zero-diagonal basis and write A_0 for the compression to their span. The other q vectors are retained as zero singleton diagonal blocks. Then $\text{tr } A_0 = 0$ and $\|A_0\| \leq a$. All off-diagonal interactions remain part of A and will be handled by the assembly lemma.

If $A_0 = 0$, every diagonal block in this decomposition is zero, and the assembly lemma applies directly. Otherwise there are two cases.

If A_0 satisfies $\text{HM}(t_0)$, Proposition 5.3 gives a representation of A_0 of cost at most $K_H \|A_0\| \leq K_H a$. Assembling this block with the q zero singletons gives cost at most $(8sK_H + b)a \leq Ka$. The condition is tested with A_0 's own norm. When $q = 0$, this step is still a direct application of the high-mass proposition; it does not appeal to induction in dimension n .

If A_0 fails $\text{HM}(t_0)$, some θ satisfies

$$\text{tr} | \text{Re}(e^{i\theta} A_0) | < t_0 N \|A_0\| = 2k \|A_0\|.$$

Proposition 6.3 supplies R trace-zero blocks T_i of dimension $k < n$, with $\|T_i\| \leq (319/R)a$. Strong induction gives commutator representations of these blocks of cost at most $(319K/R)a$. Together with the q zero singletons there are $R + q \leq 2R - 1$ blocks. Applying Lemma 7.1 gives

$$\|B\| \|C\| \leq \left(\frac{8s \cdot 319}{R} K + b \right) a = \left(\frac{2552}{8192} K + b \right) a \leq Ka,$$

because $2552/8192 < 1/2$ and $K \geq 2b$. This closes the induction. $\square$

Remark 7.2 (Size and scope of the bound). The iterate in (5.3) makes (7.2) extremely large. Its role is to establish a finite bound, not to optimize it. The number of shear operations depends only on the fixed t_0 , not on n . The final factors need not be normal. Auxiliary doubled spaces occur only in the vector-selection argument; the commutator representation itself always acts on the original space.

A Complete Lean formalization

The companion project contains a complete proof of the main theorem. Its principal declaration, in `NoEpsilon/Main.lean`, is

`NoEpsilon.uniformCommutatorBound,`

which proves the proposition `NoEpsilon.UniformCommutatorBound`. The declaration quantifies over one positive constant K , then over every dimension n and every trace-zero complex $n \times n$ matrix. Its factors are matrices of that same dimension. The Lean theorem also includes the empty dimension.

The scoped matrix norm is the Euclidean operator norm. A second audited statement spells out the norm as the norm of the continuous linear operator on Euclidean space:

`NoEpsilon.uniformCommutatorBound_euclideanOperatorNorm.`

Thus the formal result concerns operator norms, rather than entrywise or Frobenius norms.

The checked environment is Lean 4.30.0-rc1, with mathlib commit

`0c154d67103f74be3a0f2c509f72ccbf5be9f2a7.`

The complete dependency chain has proof terms, including the following steps:

- The adaptive two-commutator decomposition, the Riccati fixed point, and the identity-corner estimate $P(a)$.
- Compressed numerical ranges, neutral vectors with large images, the proportional-rank bridge, and its coordinate completion.
- Finite rectangular shear elimination with its trace and condition-number bounds, followed by the complete high-mass commutator construction.
- Two-dimensional Steinitz rearrangement and exact grouping, together with the three-Hermitian diagonal-control theorem.
- Finite complex-vector MSS selection: the mixed characteristic polynomial identity, real stability, quantitative barrier estimates, interlacing selection, covariance-deficit padding, and scaling.
- The paired selection theorem, the entire balanced quota tree, and the trace-zero low-mass paving with bound $319/R$.
- Rectangular Sylvester inversion, finite-block assembly with arbitrary first factors, all remainder coordinates, and the final dimension induction.

In particular, the Steinitz, hollowisation, and MSS inputs cited in the mathematical exposition are proved within the formal dependency chain; they are not supplied as additional mathematical axioms. Intermediate reductions with explicitly supplied branch hypotheses are retained as reusable lemmas, but those hypotheses are discharged before the principal theorem is invoked.

The project build and kernel dependency audit completed successfully. The principal theorem and its explicit Euclidean-operator-norm version depend only on

`propext, Classical.choice, Quot.sound.`

These are Lean’s standard logical axioms. The audited dependency closures contain no `sorryAx` or additional mathematical axiom. The source project records the exact commands, audit output, and source hashes in its `verification/` directory. The formalization uses some alternate proofs of intermediate results; the kernel checks its precise theorem statements and proof terms, while this manuscript supplies the informal mathematical exposition.

B Points requiring particular care in review

The logical dependencies in the proof are

Hermitian decomposition $\longrightarrow$ identity-corner core,
 compressed numerical ranges $\longrightarrow$ proportional bridge $\longrightarrow$ high-mass bound,
 Steinitz + hollowisation + paired vector selection $\longrightarrow$ low-mass paving,
 high-mass bound + low-mass paving + Sylvester assembly $\longrightarrow$ strong induction.

The rank-one argument in Lemma 5.2 has exactly three real affine constraints. The orthogonal-vector construction uses all four families $v_i, Av_i, A^*v_i, A^*Av_i$. Lemma 6.1 does not assert simultaneous constant diagonals for all three Hermitian matrices. The binary selection in Lemma 6.2 controls both complementary children and partitions indices, not merely vector values. Finally, the induction checks the retained compression against its own norm and accounts for all remainder blocks.

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